

Austin Luk

778-680-2954 | Vancouver, BC | lukaustin0@gmail.com | [linkedin.com/in/austinluk](https://www.linkedin.com/in/austinluk) | github.com/austinluk

EDUCATION

University Of British Columbia

Graduating May 2027

Bachelor of Science in Computer Science

Relevant Coursework: Machine Learning (CPSC 330), Introduction to Artificial Intelligence (CPSC 322) Data Structures and Algorithms (CPSC 221), Software Construction (CPSC 210), Database Systems (CPSC 304),

Langara College

Sept 2023 – April 2025

Associate of Science Computer Science

Activities and societies: Vice-President of Langara Volleyball Club, Langara Computer Science Club (LCC)

EXPERIENCE

Code Ninjas

March 2025 - Current

Instructor

Vancouver, BC

- Troubleshoot software, hardware, and connectivity issues in a fast-paced classroom environment, supporting 50+ students.
- Diagnosed and resolved technical problems across Windows and browser-based platforms under time constraints
- Taught programming fundamentals (Scratch, JavaScript, Lua, C#/Unity) by guiding students through debugging and problem-solving workflows

Headstarter AI

July 2024 – Sept 2024

Software Engineer Fellow

Vancouver, BC

- Debugged and fixed environment and dependency issues across full-stack applications, improving development.
- Collaborated in agile teams to build and deploy full-stack applications using MVC architecture (Next.js + backend APIs), contributing features across frontend and backend
- Designed AI-powered applications using **OpenAI APIs + vector-based retrieval (RAG)** to improve response relevance.

PROJECTS

[CourtIQ \(Volleyball Tracking System\)](#) | Python, PyTorch, OpenCV

- Developed a real-time computer vision system to track volleyball player movements and ball trajectory from video input
- Trained and **fine-tuned** object detection models in PyTorch for motion tracking
- Built a data processing pipeline to extract positional data and generate gameplay analytics (e.g., player movement, ball speed, rally patterns)
- Developed a data pipeline to extract positional data and compute metrics (player movement paths, ball speed, rally duration)

[TaPRO](#) | Gemini API, Next.js, Node.js, Supabase, AWS

- Built an AI-powered virtual teaching assistant platform to support university office hours for students and professors.
- Designed data pipelines to store assignments, course materials, and interaction summaries in Supabase for fast retrieval and reuse.
- Implemented optimized retrieval pipelines to reduce API payload size and improve response latency by 40%.

[ThenStep](#) | Next.js, Bun, OpenStreetMap, Google Maps API, Puppeteer

- Developed an advanced map application that allows users to search for both normal driving and walking routes, as well as scenic alternatives.
- Used Puppeteer for web scraping to gather data on safety reports, local meetups, and community events.
- Integrated multiple APIs (**OpenStreetMap + Google Maps**) for dynamic route rendering and interactive mapping.

TECHNICAL SKILLS

Operating Systems: macOS, Windows, Linux

Languages: Python, Java, JavaScript, TypeScript, C++, SQL, HTML, CSS,

Frameworks & Libraries: React, Node, Express, REST APIs, Git, MySQL, MS SQL, MongoDB

DevOps: Amazon Web Services (AWS), Docker, Kubernetes, Azure, CI/CD

Certifications: AWS Certified Cloud Practitioner (In Progress – Expected May 2026)